

Course Handout

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Name	Object Oriented Programming	Session	July – December 2026
Course Code	25CSE0201	Semester/Batch	3 rd /2025
L-T-P (Per Week)	2-0-4	Course Credits	4
Pre-requisite	Basic concepts of computer fundamentals	NHEQF Level ¹	5
Course Coordinator	Dr. Heena Wadhwa	SDG Number ⁴	4, 8, 9

1. Objectives of the Course

The scope of the course is to provide the foundation for understanding the key aspects of Java programming and implementation obtaining a theoretical understanding of advance programming concepts. The objectives of the course are:

- To understand core Java programming concepts including data types, control structures, methods, and arrays.
- To design and implement object-oriented solutions using classes, inheritance, polymorphism, abstraction, and interfaces.
- To utilize strings, wrapper classes, and collections framework for efficient data manipulation.
- To develop robust applications using exception handling and custom exceptions.
- To apply file handling and serialization techniques for data storage and retrieval.

2. Course Learning Outcomes (CLOs)

Students should be able to:

	CLOs	Program Outcomes (PO)	NHEQF Level Descriptor ²	No. of Lectures
CLO01	Apply core Java programming concepts including data types, control structures, methods, and arrays to solve computational problems.	PO1, PO2, PO3, PO5	Q1, Q2	19
CLO02	Design object-oriented solutions using classes, inheritance, polymorphism, abstraction, and interfaces.	PO1, PO2, PO3, PO5, PO9	Q2,Q3	17
CLO03	Implement string handling, wrapper classes, and collections framework for efficient data manipulation.	PO2, PO5	Q2,Q3	19
CLO04	Develop robust applications using exception handling mechanisms and custom exceptions.	PO1, PO2, PO3, PO5, PO10	Q2, Q3, Q4	17
CLO05	Apply file handling and serialization techniques for data storage and retrieval in Java applications.	PO1, PO3, PO4, PO5	Q2, Q3, Q4	18
Total Contact Hours				90

CLO-PO Mapping

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	Type of Assessment's ³
CLO01	H	M	L		L							H	L	Formative, Summative
CLO02	M	M	L		M				L			H	M	Formative, Summative
CLO03		L			M								H	Formative, Summative
CLO04	M	H	L		M					M		H	M	Formative, Summative
CLO05	M		M	M	L							M	L	Formative, Summative

H=High, M=Medium, L=Low

3. Recommended Books:

- B01:** Herbert Schildt, "Java the Complete Reference", 9th Edition.
B02: Kathy Sierra, "OCA/OCP Java SE 7 Programmer I & II Study Guide", Oracle Press.
B03: Introduction to Java programming, Y. Daniel Liang, Pearson Education
B04: Programming in Java, S. Malhotra, S. Chudhary, 2nd edition, Oxford Univ. Press
B05: Java Programming and Object-oriented Application Development, R. A. Johnson, Cengage Learning.

¹ National Higher Education Qualification Framework Level, Refer to annexure

² NHEQF Level Descriptor, Refer to Annexure & [Learning outcomes descriptors for qualification for all levels on the NHEQF](#)

³Types of Assessments can be referred from Type of Assessments. Refer to Annexure.

⁴For SDG Mapping with Courses, PI refer [SDG Mapping policy for Courses](#)



4. Other readings and relevant websites:

Serial No	Link of Journals, Magazines, websites and Research Papers
1.	https://onlinecourses.nptel.ac.in/e-learning/preview/noc22_cs47
2.	https://cse.iitkgp.ac.in/~dsamanta/java/index.htm
3.	https://www.w3schools.com/java/default.asp
4.	https://www.geeksforgeeks.org/java/object-oriented-programming-oops-concept-in-java/

5. Recommended Tools and Platforms

- Visual Studio IDE
- Testpad

6. Course Plan: Theory+ Lab Plan

Theory Plan

Lect. No.	Topic(s)
1-2	Detail Discussion of Course Handout(CHO) Java introduction, JDK, JRE, JVM, Data types, variables, operators: Overview of Java platform and fundamental programming concepts including data representation and operations.
3-4	Input/output and comments, Control statements and loops for performing basic I/O operations and implementing decision-making and iterative approach.
5-7	Methods and parameter passing, Arrays and strings basics for developing modular programs and handling data using arrays and basic string operations.
8-9	Classes and objects, Constructors and this keyword for applying object-oriented concepts for creating and initializing objects effectively.
Continuous Evaluation1 (1-9)	
10-11	Static members and blocks: Understanding class-level variables, methods, and static initialization mechanisms.
12-13	Inheritance, Polymorphism, Abstraction and Interfaces
13-15	String class and string methods, StringBuilder and string comparisons for performing advanced string manipulation and efficient string handling.
16-19	Multidimensional arrays, Packages and access modifiers for organizing data structures and controlling access in large-scale applications.
Continuous Evaluation 2 (10-19)	
20-21	Wrapper classes and autoboxing, Revision and practice
22-23	Exception hierarchy, try, catch, finally, throw and throws for handling runtime errors using structured exception handling techniques.
24	Custom exceptions, File handling with streams, Serialization basics
25	Collection framework overview: Understanding the architecture and components of Java Collections Framework
26-27	ArrayList and LinkedList: Working with ordered and dynamic collections for efficient data management.
28	HashSet and TreeSet for managing unique data elements using unordered and sorted set structures.
Continuous Evaluation 3 (20-28)	
29	HashMap and TreeMap for storing and processing key-value pairs using hashing and sorting techniques.
30	Iterators and loops over collections, Comparators and sorting collections
End Term Examination (1-30)	

Lab Plan

Lab No.	Topic(s)
1-2	Implement a program and demonstrate basic Java structure. Demonstrate use of data types, variables, and operators.
3-4	Implement input and output operations using Scanner and System.out.
5-7	Write a program to demonstrate use of conditional statements (if-else, switch).
8-10	Implement different looping constructs (for, while, do-while).
11-12	Write a program to demonstrate method creation and parameter passing.
13-16	Implement array operations like sum, average, and searching.
17-19	Write a program to perform basic string operations (length, substring, concat)
20-23	Implement a program to demonstrate classes and objects.
24-27	Write a program to demonstrate constructors and use of this keyword.
28-30	Implement a program to demonstrate static variables and static methods.
31-32	Implement inheritance using single inheritance. Write a program to demonstrate multilevel inheritance.
33-34	Implement method overloading (compile-time polymorphism). Write a program to demonstrate method overriding (runtime polymorphism).
35-38	Implement abstraction using abstract classes. Write a program to demonstrate interfaces.
39-42	Implement string manipulation using String class methods. Write a program using StringBuilder for efficient string operations.
41-43	Write a program for multidimensional arrays (matrix addition/multiplication). Write a program to demonstrate packages and access modifiers. Implement wrapper classes

44-48	Implement exception handling using try, catch, and finally. Write a program using throw and throws keywords. Implement custom (user-defined) exceptions.
49-52	Write a program for file handling using FileInputStream and FileOutputStream. Implement serialization and deserialization of objects.
53-55	Implement ArrayList operations (add, remove, search). Write a program to demonstrate LinkedList operations.
56-58	Implement HashSet and TreeSet operations. Write a program to demonstrate HashMap and TreeMap operations.
59-60	Write a program to sort collections using Comparable interface. Implement sorting using Comparator interface.

7. Delivery/Instructional Resources Theory plan:

Lect No.	Topics	CLO	Book No, CH No, Page No	TLM ³	ALM ⁴	Web References	Audio-Video
1-2	Java introduction, JDK, JRE, JVM, Data types, variables, operators	1	B01, CH 1, B01, CH 2	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.oracle.com/webfolder/technetwork/tutorials/obe/java/gc01/index.html	https://nptel.ac.in/courses/106105191
3-4	Input/output and comments, Control statements and loops	1	B03, CH 2 B02, CH 2	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.geeksforgeeks.org/decision-making-javaif-else-switch-break-continue-jump/ https://www.geeksforgeeks.org/command-line-arguments-in-java/	https://youtu.be/FEkceTEI5DA
5-7	Methods and parameter passing, Arrays and strings basics	1	B03, CH 6 B03, CH 10	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.w3schools.com/java/java_methods_parameters.asp	https://www.youtube.com/watch?v=qcSz4ef9UHA
8-9	Classes and objects, constructors and this keyword	1,2	B04, CH 3 B05, CH 4	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.w3schools.com/java/java_constructors.asp https://www.geeksforgeeks.org/classes-objects-java/	https://nptel.ac.in/courses/106105191
10-11	static members and blocks, Inheritance, Polymorphism	2	B01, CH 6 B02, CH 6 B05, CH 5	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.w3schools.com/java/java_inheritance.asp https://www.geeksforgeeks.org/inheritance-in-java/	https://nptel.ac.in/courses/106105191
12-13	Abstraction and interfaces	1	B01, CH 8 B01, CH 9	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.w3schools.com/java/java_abstract.asp https://www.geeksforgeeks.org/packages-in-java/	https://nptel.ac.in/courses/106105191
13-15	String class and string methods, StringBuilder and string comparisons	1, 2	B03, CH 10	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.w3schools.com/java/java_ref_string.asp https://www.geeksforgeeks.org/stringbuffer-class-in-java/?ref=lbp	https://www.youtube.com/watch?v=3xuJlaP3C4g https://www.youtube.com/watch?v=3xuJlaP3C4g
16-19	Multidimensional arrays	1,2	B03, CH 7	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.w3schools.com/java/java_arrays_multid.asp	https://www.youtube.com/watch?v=18Zt5I4S45o
20-21	Packages and access modifiers, Wrapper classes and autoboxing	3	B01, CH 9	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.geeksforgeeks.org/java/autoboxing-unboxing-java/	https://www.youtube.com/watch?v=Fyc86kVlePE

³ Teaching Learning Methods, Refer to Annexure

⁴ Active Learning Methods

22-23	Exception hierarchy, try, catch, finally, throw and throws and Custom exceptions	4,5	B01, CH 10 B04, CH08	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.w3schools.com/java/java_try_catch.asp https://www.geeksforgeeks.org/exceptions-in-java/?ref=lbp	https://nptel.ac.in/courses/106105191
24	File handling with streams, Serialization basics	4,5	B01, CH 11	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.geeksforgeeks.org/java/serialization-and-deserialization-in-java/	https://www.youtube.com/watch?v=Uxhb3IHKMQk
25	Collection framework overview	4,5	B01, CH 18	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge BAssignment	https://www.javatpoint.com/collections-in-java https://www.geeksforgeeks.org/collections-in-java-2/	https://www.youtube.com/watch?v=GdAon80-0KA
26-27	List: ArrayList and LinkedList	1,3	B02, CH 05	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.geeksforgeeks.org/java/arraylist-vs-linkedlist-java/	https://www.youtube.com/watch?v=liFyhZl9uw
28	Set: HashSet and TreeSet	3	B01, CH18	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.geeksforgeeks.org/java/hashset-vs-treeset-in-java/	https://www.youtube.com/watch?v=plcZvG_TeAk
29	Map: HashMap and TreeMap	3	B01, CH18	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.geeksforgeeks.org/java/hashmap-treemap-java/	https://www.youtube.com/watch?v=WeF3_nk-UqY
30	Iterators and loops over collections Comparators and sorting collections	5	B03, CH21 B02, CH05	Lecture Discussion Demonstration method using a simulation or a tool	Quiz Lab Challenge Assignment	https://www.geeksforgeeks.org/java/iterators-in-java/	https://www.youtube.com/watch?v=NRrNw2XoVB8

Lab Plan:

Lab No.	Experiment	CLO	TLM	ALM	Web References	Audio-Video
1-2	Implement a program and demonstrate basic Java structure. Demonstrate use of data types, variables, and operators.	1	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=eIrMbAQSU34
3-4	Implement input and output operations using Scanner and System.out.	1	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=UmnCZ7-9yDY
5-7	Write a program to demonstrate use of conditional statements (if-else, switch).	1	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=ldYLYI
8-10	Implement different looping constructs (for, while, do-while).	1	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://youtu.be/FEkceTEI5DA
11-12	Write a program to demonstrate method creation and parameter passing.	1,2	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=qcSz4ef9UHA
13-16	Implement array operations like sum, average, and searching.	1,2	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=aSGa3S2-sQ0
17-19	Write a program to perform basic string operations (length, substring, concat)	1,3	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=n76nBLwIwRo
20-23	Implement a program to demonstrate classes and objects.	1,2	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=BSVKUk58K6U

24-27	Write a program to demonstrate constructors and use of this keyword.	2	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=8cm1x4bC61
28-30	Implement a program to demonstrate static variables and static methods.	2	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://nptel.ac.in/courses/106105191
31-32	Implement inheritance using single inheritance. Write a program to demonstrate multilevel inheritance.	2	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://nptel.ac.in/courses/106105191
33-34	Implement method overloading (compile-time polymorphism). Write a program to demonstrate method overriding (runtime polymorphism).	1,2	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://nptel.ac.in/courses/106105191
35-36	Implement abstraction using abstract classes. Write a program to demonstrate interfaces.	2,3	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://nptel.ac.in/courses/106105191
37-38	Implement string manipulation using String class methods. Write a program using StringBuilder for efficient string operations.	3	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=3xuJlaP3C4g
39-41	Write a program for multidimensional arrays (matrix addition/multiplication). Write a program to demonstrate packages and access modifiers. Implement wrapper classes	2,3	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=18Zt5I4S45o
42-48	Implement exception handling using try, catch, and finally. Write a program using throw and throws keywords. Implement custom (user-defined) exceptions.	2,4,5	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://nptel.ac.in/courses/106105191
49-52	Write a program for file handling using FileInputStream and FileOutputStream. Implement serialization and deserialization of objects.	4,5	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=Uxhb3IHKMQk
53-56	Implement ArrayList operations (add, remove, search). Write a program to demonstrate LinkedList operations.	3	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=liFyhZl9uw
57-58	Implement HashSet and TreeSet operations. Write a program to demonstrate HashMap and TreeMap operations.	3,5	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=plcZvG_TeAk
59-60	Write a program to sort collections using Comparable interface. Implement sorting using Comparator interface.	2,3	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=NRrNw2XoVB8

8. Remedial Classes⁵

After every continuous evaluation, identify weak learners, and prepare the student lists. The following methods would be used to improve the performance of students.

- Remedial Classes would be planned and scheduled accordingly.
- Offer supplementary materials or activities to reinforce concepts.

9. Self-Learning⁶

Assignments to promote self-learning, survey of contents from multiple sources.

S.No	Topics	CLO	ALM	References/MOOCs
1	Linear search implementation	1, 6	Think – Pair- Share	https://www.geeksforgeeks.org/dsa/linear-search/
2	Tic-Tac-Toe Game	1,2,6	Think – Pair- Share	https://www.geeksforgeeks.org/java/tic-tac-toe-game-in-java/
3	Find the nth Fibonacci Number	1,6	Think – Pair- Share	https://www.geeksforgeeks.org/dsa/program-for-nth-fibonacci-number/

⁵ Refer to Annexure

⁶ Refer to Annexure

10. Delivery Details of Content Beyond Syllabus⁷

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.No	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCs
1	Multithreading in Java	6	1, 2, 4	Brainstorming session	https://www.geeksforgeeks.org/java/multithreading-in-java/

11. Evaluation Scheme & Components:

Assessment Type	Evaluation Component	Type of Component	No. of Assessments	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative	Component 1	Continuous Evaluation 1	01*	20%	20	Computer based online in Campus	1-5
Formative	Component 2	Continuous Evaluation 2 & 3	02*	25%	50	Computer based online in Campus	1-5
Summative	Component 3	End Term Examination	01**	30%	30	Computer based online in Campus	1-5
Total			100%				

* Students will have to appear in all evaluations.

**As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

12. Syllabus of the Course:

Subject: Object Oriented Programming			
S.No.	Topic(s)	No. of Lectures	Weightage %
1	Java introduction, JDK, JRE, JVM, Data types, variables, operators: Overview of Java platform and fundamental programming concepts including data representation and operations. Input/output and comments, Control statements and loops for performing basic I/O operations and implementing decision-making and iterative approach. Methods and parameter passing, Arrays and strings basics for developing modular programs and handling data using arrays and basic string operations. Classes and objects, Constructors and this keyword for applying object-oriented concepts for creating and initializing objects effectively.	32	35%
Continuous Evaluation 1			
2	Static members and blocks: Understanding class-level variables, methods, and static initialization mechanisms. Inheritance, Polymorphism, Abstraction and Interfaces String class and string methods, StringBuilder and string comparisons for performing advanced string manipulation and efficient string handling. Multidimensional arrays, Packages and access modifiers for organizing data structures and controlling access in large-scale applications. Wrapper classes and autoboxing, Revision and practice Exception hierarchy, try, catch, finally, throw and throws for handling runtime errors using structured exception handling techniques.	27	30%
Continuous Evaluation 2			
3	Custom exceptions, File handling with streams, Serialization basics Collection framework overview: Understanding the architecture and components of Java Collections Framework ArrayList and LinkedList: Working with ordered and dynamic collections for efficient data management. HashSet and TreeSet for managing unique data elements using unordered and sorted set structures.	25	30%
Continuous Evaluation 3			
4	HashMap and TreeMap for storing and processing key-value pairs using hashing and sorting technique, Iterators and loops over collections, Comparators and sorting collections	06	5%
End Term Examination			

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

⁷ Refer to Annexure

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Heena Wadhwa	
Head-Academic Delivery	Dr. Mrinal Paliwal	
Dean	Dr. Rishu Chhabra	
Date (DD/MM/YYYY)	12.6.2026	

Annexure

1. Pre- requisite

Mention The Pre-requisite skill set or course/s if it is expected to be studies before this course, otherwise write “not applicable”.

2. NHEQF levels

The NHEQF levels represent a series of sequential stages expressed in terms of a range of learning outcomes against which typical qualifications are positioned/located. NHEQF level 4.5 represents learning outcomes appropriate to the first year (first two semesters) of the undergraduate programme of study, while Level 8 represents learning outcomes appropriate to the doctoral-level programme of study.

Table 1: Higher education qualifications at different levels on the NHEQF

NHEQF level	Examples of higher education qualifications located within each level
Level 4.5	Undergraduate Certificate. Programme duration: First year (first two semesters) of the undergraduate programme, followed by an exit 4-credit skills-enhancement course(s).
Level 5	Undergraduate Diploma. Programme duration: First two years (first four semesters) of the undergraduate programme, followed by an exit 4-credit skills-enhancement course(s) lasting two months.
Level 5.5	Bachelor's Degree. Programme duration: First three years (Six semesters) of the four-year undergraduate programme.
Level 6	Bachelor's Degree (Honours/ Honours with Research). Programme duration: Four years (eight semesters).
Level 6	Post-Graduate Diploma. Programme duration: One year (two semesters) for those who exit after successful completion of the first year (two semesters) of the 2-year master's programme.
Level 6.5	Master's degree. (e.g. M.A., M.Com., M.Sc., etc.) Programme duration: Two years (four semesters) after obtaining a 3- year Bachelor's degree (e.g. B.A., B.Sc., B.Com.etc.).
Level 6.5	Master's degree. (e.g. M.A., M.Com., M.Sc., etc.) Programme duration: One year (two semesters) after obtaining a 4 -year Bachelor's degree (Honours/ Honours with Research) (e.g. B.A., B.Sc., B.Com. etc.).
Level 7	Master's degree.(e.g. M.E./M.Tech. etc.) Programme duration: Two years (four semesters) after obtaining a 4-year Bachelor's degree. (e.g. B.E./B.Tech. etc.)
Level 8	Doctoral Degree

3. NHEQF level descriptors

Each NHEQF level is structured based on the defined learning outcomes which lead to the expected graduate attributes/profile. The level descriptors reflect the expected outcomes of learning that should be achieved and demonstrated by graduates of a specific programme of study leading to a qualification at a specific NHEQF level.

Click [Learning outcomes descriptors for qualification for all levels on the NHEQF](#)

4. Course Outcomes

The number of Course Outcomes is recommended to be 4-5 for courses that do not contain practical component and 6 for those courses with a practical component. Flexibility can be sought by the post-graduate courses in this regard.

5. Theory/lab Plan

The following are the guidelines to be followed while creating plans

- Each session may be planned for a duration of 45/50mins (irrespective of the double hour or single hour scheduled in timetable).
- Every session must incorporate at least one active learning method which may or may not be part of the assessments.
- Put BoS Approved Syllabus in the topics. Deviations (if any) from BoS approved syllabus must be brought to the notice of BoS chairman & Dean Academics, After approval, revised handout should be submitted.
- The Topics elaborated in the Theory/Lab plan must match those in the course execution plan.

6. Teaching Learning Methods

The following are some of the Teaching & Learning methods that can be incorporated in session wise teaching learning plan.

- **Teacher-centered Learning Methods:**
 - Lecture
 - Discussion
 - Demonstration method using a simulation or a tool
 - Reviewing
 - Questioning
- **Learner-centered teaching & Learning methods:**

- i. **Active learning**, in which students solve problems, answer questions, formulate questions of their own, discuss, explain, debate, or brainstorm during class;
- ii. **Cooperative learning**, in which students work in teams on problems and projects under conditions that assure both positive interdependence and individual accountability; and
- iii. **Inductive teaching and learning**, in which students are first presented with challenges (questions or problems) and learn the course material in the context of addressing the challenges.
- iv. **Inductive methods** include inquiry-based learning, case based instruction, problem-based learning, project-based learning, discovery learning, and just-in-time teaching. It is important to integrate authentic, reflective and collaborative learning experiences when designing for student-centered learning.

7. Active Learning Methods

The following are some of the Active Learning Methods that can be incorporated in session wise teaching learning plan.

- One Minute Paper
- Group Discussion
- Student-Created PPT, Charts, Matrices, Flowcharts, Models
- The Fish Bowl
- Debate
- Video Synthesis
- Quiz/Test Questions
- Brain Storming Sessions
- Case Study
- Shadowing
- Leading Question
- Puzzle, Enigma, Contradiction
- Statement-Opinion-Summary
- Think / Pair / Share
- Peer Review
- Just in Time Teaching
- Statement-Opinion-Summary
- Peer Survey
- Focused Listing
- Role-Playing
- Student Field Work with Reflection
- Infusing Humor into Class Sessions
- Inviting Effective Guest Speakers

8. Remedial Classes

After every Sessional Test, identify weak learners, provide supplement course handout. Student list and Impact Observed report should be submitted to Dean through proper channel.

9. Self Learning

Plan 10% of topics in self-learning mode with discussions, ALM's and Assessment happening in the class.

10. Content Beyond Syllabus

Plan Advanced Topics, Experiments, Additional Reading, Research papers in self-learning mode with ALM's and Assessment happening in the regular class or lab. Usually caters advanced learners. Identify Advanced learners. For Extra classes, schedule should be notified accordingly.

11. Assessment Type

1. Assessment broadly can be classified into the following types:

- a. **Diagnostic assessments:** Diagnostic assessments are intended to help teachers identify what students know and can do in different domains to support their students' learning. These help teachers determine strengths of students in various areas to better address their specific needs.
- b. **Formative assessments:** Formative assessment refers to a wide variety of methods that teachers use to conduct in-process evaluations of student comprehension, learning needs, and academic progress during a lesson, unit, or a course. Formative assessments help teachers identify concepts that students are struggling to understand, skills they are having difficulty acquiring, or learning standards they have not yet achieved so that adjustments can be made to lessons, instructional techniques, and academic support.
- c. **Summative assessments:** Summative assessment is an assessment administered at the end of an instructional unit in a course. These assessments are intended to evaluate student learning by comparing performance to a standard or benchmark.
- d. **Ipsative assessments:** Ipsative assessment involves comparisons between past and current work to identify a learner's growth over time, rather than progress toward an external set of criteria. Therefore, Ipsative assessment is an internal or self-referenced assessment.
- e. **Norm-referenced assessments:** Norm-referenced tests report whether test takers performed better or worse than a hypothetical average student, which is determined by comparing scores against the performance results of a statistically selected group of test takers, typically of the same age or grade level, who have already taken the exam.
- f. **Criterion-referenced assessments:** Criterion-Reference tests measure the performance of test takers against the criteria covered in the curriculum.
- g. **Peer-to-Peer randomised Assessments:** Peers will be able to provide assessment in this case
- h. **Industry Validation of Effectiveness:** In the Vocation Education, Industry validation of effectiveness of training is particularly important.
- i. **Self-assessments:** To evaluate how much the learner has grasped by self-learning.

2. Other Assessment Methods: Conducting an assessment takes time, thought, attention, planning, and often collaboration. Each assessment tool, whether a short survey or detailed rubric, will be useful only insofar as it both addresses the outcomes well and is feasible to use.

a. **Rubrics:** For assessing qualitative student work such as essays, projects, reports, or presentations. Rubrics serve well to clearly denote the specific expectations for an assignment, for collecting data for assessment of student learning outcomes, and for student performance. Rubrics can be used for grading, for providing feedback to students, and for informing and encouraging students to think about their own learning.

b. **Portfolios and E-Portfolio:** Portfolios can provide a window into the process of student learning across a semester-long project that can be assessed (usually by using a rubric).

c. **Curriculum Mapping:** A good curriculum map can serve to focus assessment, and the improvements that follow, where it will be most useful, informative, or effective.

d. **Structured Interviews:** While time-consuming, structured interviews are useful when specific questions need to be asked. It also leaves room for unplanned topics or ideas to emerge.

e. **Student Experience Surveys:** Student experience in research universities (SERU), including administration of on-line census SERU Undergraduate and Graduate Surveys, can yield important information about student perceptions and experiences.

12. Evaluation Component & Types

As per LMs we need to figure it out whether it is component 1, 2 or 3. In Types of Evaluation Component, we need to specify what type of evaluation we are performing like Continuous Evaluation or Sessional Test or End Term Examination.

13. No. of Assessments and Weightage of Components

Department will give guideline for number of assessments, mandatory or optional and weightage.